Sagar Kriplani

Mechanical Design & Development Engineer

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Mechanical design engineer with experience in special-purpose machinery, semiconductor equipment and scientific instrumentation. Combines manufacturable CAD design, production documentation and experimental thermal investigation.

EXPERIENCE

NeSt GmbH | Mechanical Design Engineer · Konstrukteur

09/2025 - 09/2026 | Herbrechtingen, Germany

Developed machine enclosures, precision assemblies, sheet-metal details and manufacturing drawings using Autodesk Inventor, with DFM/DFA and production coordination. Project figures reported by me: 8+ machine platforms, approximately 200 drawings per month, component cost reductions up to 70% and weight reduction of 60%. These figures are not independently established by the employer reference.

Brooks Automation | Mechanical Design Engineer (Working Student)

02/2024 - 06/2024 | Steißlingen, Germany

Designed precision components and assemblies for semiconductor automation, developed CAD models and manufacturing drawings, and collaborated with R&D, electrical engineering and production.

Brooks Automation | Research & Development · Master's Thesis

07/2023 - 01/2024 | Steißlingen, Germany

Compared radiant heaters and LEDs for polymer heating in vacuum chambers. Developed thermal models and experimental setups to assess heating performance and thermal uniformity. The thesis was submitted separately in February 2025.

Brooks Automation | Engineering Intern · Mechanical Design

01/2023 - 06/2023 | Steißlingen, Germany

Analysed the Process Air Unit, compared heater concepts and developed mechanical design and assembly documentation for semiconductor automation equipment. The issued internship reference reports 20% cost reduction; the project report separately records approximately 21%.

Ravi Brass Industries | Design & Development Engineer

08/2020 - 09/2021 | Jamnagar, India

Designed mechanical components and assemblies, prepared manufacturing drawings and coordinated with production on manufacturability and quality.

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PROJECTS & QUALIFICATIONS

PAU redesign · Brooks internship

Thermodynamic calculations, component selection and mechanical design. Project report: approximately 500 to 50 mbar pressure drop and 48 to 18 kW heater design capacity; not measured operating energy savings.

Polymer heating · Master's thesis

Compared radiant heaters and LEDs in vacuum chambers. LEDs improved uniformity in the tested setup; radiant heaters performed better overall. Company research: 07/2023-01/2024; thesis submitted 02/2025.

Academic CAD & simulation

Cylindrical handling mechanism in Inventor; ANSYS training exercises; MATLAB finite-difference frost-growth model in a four-student team. Academic results are not production validation.

EDUCATION

M.Sc. Scientific Instrumentation | 10/2021 - 03/2025

Ernst-Abbe-Hochschule Jena · Grade 1.8 · Thesis 1.3 · 120 ECTS · ECTS B

B.E. Mechanical Engineering | 2017 - 2020

A. D. Patel Institute of Technology / GTU · CGPA 8.14/10 · First Class with Distinction

Diploma in Mechanical Engineering | 2014 - 2017

Government Polytechnic Jamnagar / GTU · CGPA 9.50/10

TECHNICAL SKILLS

Autodesk Inventor | Siemens NX | PTC Creo | ANSYS | MATLAB | DFM/DFA | BOM | SAGE ERP | Agile PLM

TRAINING

ANSYS Basic to Professional (2020); Siemens Centre of Excellence: NX, NC, Drafting, Robotics (2016-2017); LinkedIn Learning: Siemens NX, Engineering Drawings, Semiconductor Manufacturing (2026).

LANGUAGES

German: professional working; English: full professional; Hindi, Gujarati, Sindhi: native/bilingual. Self-reported; German certificate: B1-plus coursework.

Redacted certificates, references and transcripts: sagar-kriplani.github.io/en/evidence/